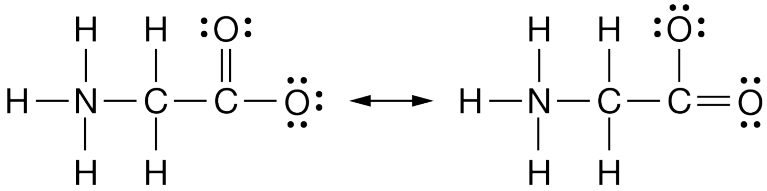
2.6

1.



Based on the resonance structures shown above, what are the bond orders of the two carbon-oxygen bonds?

(A) 1 and 1.5

(B) 1 and 2

(C) 1.5 and 1.5 ✓

(D) 2 and 2

Answer C

Correct. When two resonance structures are equivalent, they contribute equally to the average structure, and the average structure is the preferred representation of the molecule. Bond orders are obtained by taking the average of the bond orders in the resonance structures. Therefore, in this case both $C-O$ bonds have bond orders equal to the average of 1 and 2, or 1.5.

2. For which of the following molecules are resonance structures necessary to describe the bonding satisfactorily?

(A) SO_2 ✓

(B) CO_2

(C) OF_2

(D) PF_3

Directions: Each set of lettered choices below refers to the numbered statement immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

Refer to the following gaseous molecules:

(A) $BeCl_2$

(B) SO_2

(C) N_2

(D) F_2

2.6

3. Is best represented by two or more resonance forms

(A) $BeCl_2$

(B) SO_2 ✓

(C) N_2

(D) F_2

A solution is prepared by combining $1.0\ \text{L}$ of $0.50\ M\ \text{HClO}_2$ with $1.0\ \text{L}$ of $0.50\ M\ \text{KClO}_2$. The pH of the solution is 2.0.

4. Consider the Lewis electron-dot diagram for the ClO_2^- ion that minimizes formal charge. Which of the following is true about the hybridization of the Cl atom in the ion?

(A) The Cl atom is not hybridized, and the ion contains one single $\text{Cl}-\text{O}$ bond and one double $\text{Cl}-\text{O}$ bond.

(B) The Cl atom is sp hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ double bonds.

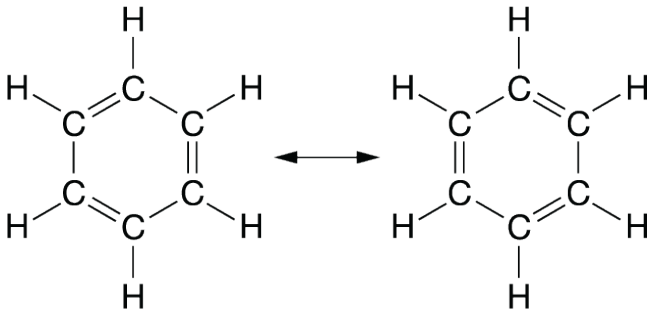
(C) The Cl atom is sp^2 hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ single bonds.

(D) The Cl atom is sp^3 hybridized, and the ion contains two equivalent $\text{Cl}-\text{O}$ single bonds. ✓

Answer D

Correct. A valid Lewis electron-dot diagram of the ClO_2^- ion has two nonbonding pairs of electrons around the central Cl atom. Since there are four electron domains around the Cl atom, it is sp^3 hybridized.

5.



The diagram above shows two resonance structures for a molecule of C_6H_6. The phenomenon shown in the diagram best supports which of the following claims about the bonding in C_6H_6 ?

2.6

(A) In the C_6H_6 molecule, all the bonds between the carbon atoms have the same length. ✓

(B) Because of variable bonding between its carbon atoms, C_6H_6 is a good conductor of electricity.

(C) The bonds between carbon atoms in C_6H_6 are unstable, and the compound decomposes quickly.

(D) The C_6H_6 molecule contains three single bonds between carbon atoms and three double bonds between carbon atoms.

Answer A

Correct. Because of resonance, all the bonds between carbon atoms in C_6H_6 are equivalent and have the same strength and equal lengths.

6.

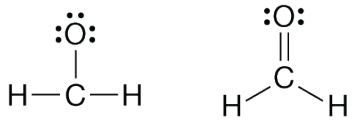


Diagram 1 Diagram 2

Which of the diagrams above best represents the CH_2O molecule, and why?

(A) Diagram 1, because all bond angles are 180°.

(B) Diagram 1, because all atoms have a formal charge of 0.

(C) Diagram 2, because the molecule has a trigonal pyramidal shape.

(D) Diagram 2, because all atoms have a formal charge of 0. ✓

Answer D

Correct. Diagram 2 represents a molecule in which both C and O have a complete octet and all atoms have a formal charge of zero.

2.6

7.

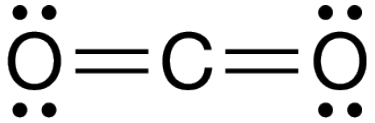


Diagram 1

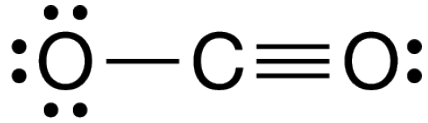


Diagram 2

Which of the Lewis diagrams shown above is the more likely structure of CO_2, and why?

(A) Diagram 1, because all the atoms have a formal charge of 0. ✓

(B) Diagram 1, because double bonds are stronger than triple bonds.

(C) Diagram 2, because all the atoms have a formal charge of 0.

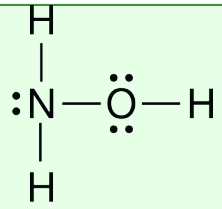
(D) Diagram 2, because triple bonds are stronger than double bonds.

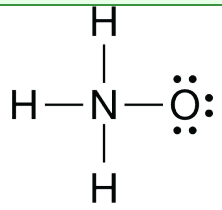
Answer A

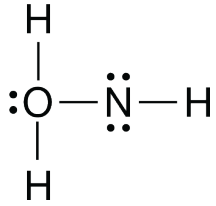
Correct. When the formal charge of each atom in diagram 1 is determined using the rule $\left(\text{formal charge of an atom} = \text{the number of valence electrons of the atom} - \left(\text{the number of electrons in lone pairs} + \text{half the number of electrons in bonds with other atoms}\right)\right)$, the results are zero for each of the three atoms. A molecular structure in which all formal charges are zero is preferable to one in which some formal charges are not zero. And in diagram 2, the results for the three atoms (going left to right) are -1, 0, and $+1$, respectively.

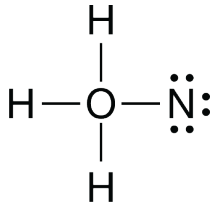
8. Based on formal charges, which of the following is the best Lewis electron-dot diagram for H_3NO?

2.6

(A)  ✓

(B) 

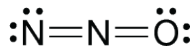
(C) 

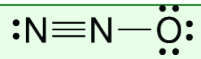
(D) 

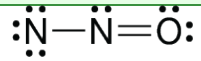
Answer A

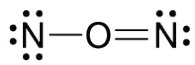
Correct. When the formal charge of each atom in this diagram is determined using the rule (formal charge of an atom = the number of valence electrons of the atom − (the number of electrons in lone pairs + half the number of electrons in bonds with other atoms)), the sum of the formal charges on all the atoms in H_3NO is equal to zero. A sum of formal charges on all atoms equal to zero is preferable.

9. Which of the following Lewis diagrams best represents the bonding in the N_2O molecule, considering formal charges?

(A) 

(B)  ✓

(C) 

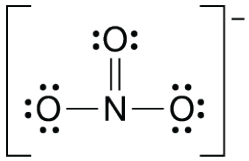
(D) 

2.6

Answer B

Correct. When the formal charge of each atom in this diagram is determined using the rule: $(\text{formal charge of an atom} = \text{the number of valence electrons of the atom} - (\text{the number of electrons in lone pairs} + \text{half the number of electrons in bonds with other atoms}))$. The sum of the formal charges on all the atoms in N_2O is equal to zero. The most favored structure considers the minimization of the formal charges and the electronegativities of the atoms. Oxygen is more electronegative than nitrogen; therefore, the Lewis diagram that shows a negative formal charge on the oxygen atom is the best for depicting the molecule.

10.



Which of the following statements, if true, would support the claim that the NO_3^- ion, represented above, has three resonance structures?

(A) The NO_3^- ion is not a polar species. ✓

(B) The oxygen-to-nitrogen-to-oxygen bond angles are 90°.

(C) One of the bonds in NO_3^- is longer than the other two.

(D) One of the bonds in NO_3^- is shorter than the other two.

Answer A

Correct. If the NO_3^- ion did NOT have three equivalent structures, then the ion would be polar because of the unequal distribution of charge within the ion. Equal contributions of the three resonance forms to the structure of the ion result in a planar ion with three equivalent polar nitrogen-oxygen bonds. Due to the trigonal planar symmetry of the NO_3^- ion, the negative dipoles cancel out.

11. Which of the following molecules has an angular (bent) geometry that is commonly represented as a resonance hybrid of two or more electron-dot structures?

(A) CO_2

(B) O_3 ✓

(C) BeF_2

(D) OF_2

2.6

12. Resonance is most commonly used to describe the bonding in molecules of which of the following?

(A) CO_2

(B) O_3 ✓

(C) H_2O

(D) SF_6